

Problem 1 [25 points] Neural Networks

Consider the neural network with one hidden layer shown in Figure 1.

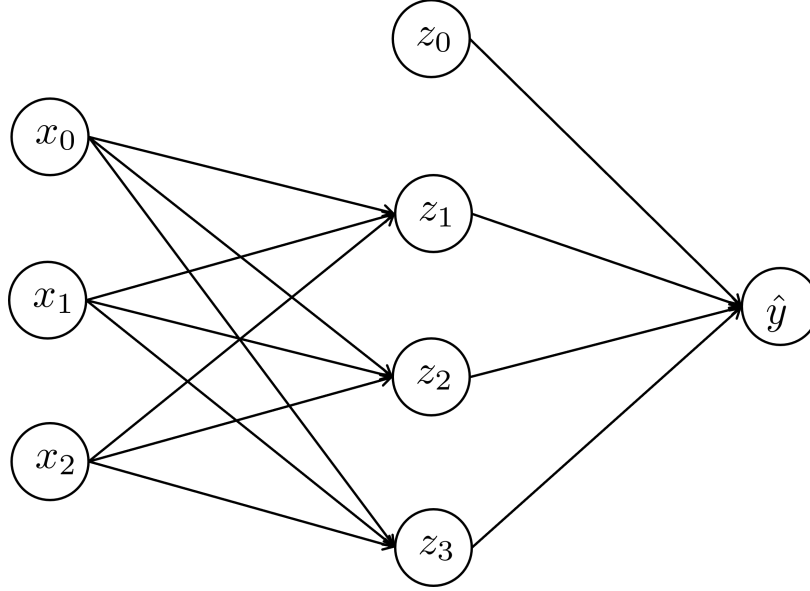


Figure 1: Fully-Connected Neural Network

The input layer consists of 2 features $x = [x_1, x_2]^T$, the hidden layer has 3 nodes with output $z = [z_1, z_2, z_3]^T$, and the output layer is a scalar $\hat{y}$. There is a bias term for each node.

α is the matrix of weights from the inputs to the hidden layer and β is the matrix of weights from the hidden layer to the output layer. Each row in the matrix represents the parameters of a neuron. For example, $\alpha_{0,0}$ is the bias in neuron z_1 , $\alpha_{0,1}$ is the weight from x_1 to z_1 , $\alpha_{0,2}$ is the weight from x_2 to z_1 , and β_i is similarly defined where β_1 is the weight from z_1 to y . We will use a tanh activation function for the hidden layer and no activation for the output layer.

$$\alpha = \begin{bmatrix} \alpha_{0,0} & \alpha_{0,1} & \alpha_{0,2} \\ \alpha_{1,0} & \alpha_{1,1} & \alpha_{1,2} \\ \alpha_{2,0} & \alpha_{2,1} & \alpha_{2,2} \end{bmatrix} \quad (1)$$

Forward Propagation is the process of calculating the value of your loss function, given data, weights and activation functions. Given the input data x , we can transform it by the given weights, α , then apply the corresponding activation function to it and finally pass the result to the next layer. Forward propagation does not involve taking derivatives and proceeds from the input layer to the output layer. In detail, we define each layer of the network as follows.

The input:

$$\mathbf{x} = [x_0, x_1, x_2]^T, \text{ where } x_0 = 1 \quad (2)$$

Linear combination at the first (hidden) layer for j -th node:

$$v_j = \sum_{i=0}^2 \alpha_{j,i} \cdot x_i \quad (3)$$

Activation at the first (hidden) layer:

$$z_j = \tanh(v_j) = \frac{e^{v_j} - e^{-v_j}}{e^{v_j} + e^{-v_j}} \quad (4)$$

$$\mathbf{z} = [z_0, z_1, z_2, z_3]^T, \text{ where } z_0 = 1 \quad (5)$$

Linear combination at the second (output) layer, we don't have activation for output layer:

$$\hat{y} = \sum_{j=0}^3 \beta_j \cdot z_j \quad (6)$$

We will use Squared error loss for each data point, $l(\hat{y}, y)$:

$$l(\hat{y}, y) = \frac{1}{2}(\hat{y} - y)^2 \quad (7)$$

We initialize the network weights α as:

$$\alpha = \begin{bmatrix} 0 & 1 & 2 \\ 2 & 1 & 0 \\ 0 & 2 & 0 \end{bmatrix} \quad (8)$$

where each rows means the weights of a node z in hidden layer (e.g., the first row $[0, 1, 2]$ means the weights for x_0, x_1, x_2 in first hidden layer neuron z_1). The network weights β is:

$$\beta = [0 \quad 1 \quad 2 \quad 2] \quad (9)$$

where each number means the weights of a hidden layer neuron z (e.g., the second number 1 indicates the weight of z_1 in output layer).

1 [5 points]. Why and how do we include a bias term in the input and in the hidden-layer?

Similar to how an intercept term in linear regression allows it to better fit data, the bias term helps the neural network better fit its data as well.

2 [5 points]. Why do we need to use nonlinear activation functions in our neural net?

A neural network with only linear activation functions would be no different than a linear regression. (Try forward propagating with only linear functions on the given example)

3 [15 points]. Given input data $x_1 = 1, x_2 = 2$, what are the values of v_1, v_2 , and v_3 ?

$$v_1 = \sum_{i=0}^2 \alpha_{1,i} x_i = 5 \quad z_1 = 0.99991 \quad \hat{y} = 4.91807$$

$$v_2 = \sum_{i=0}^2 \alpha_{2,i} x_i = 3 \quad z_2 = 0.99505 \quad l = 1.83950$$

$$v_3 = \sum_{i=0}^2 \alpha_{3,i} x_i = 2 \quad z_3 = 0.96403$$

Problem 2 [25 points] CNN

Consider the following 4×4 image and 2×2 filter below.

1	3	-2	4
0	8	6	5
2	1	-9	0
4	-1	3	7

1	2
-2	-1

1 [5 points]. Assume that there is no padding and stride = 1. What are the dimensions of the output, and what is the value in the bottom right corner of the output image?

Output is 3x3, and the bottom right value is $-9 + 0 - 6 - 7 = -22$.

2 [5 points]. Now assume that we having padding = 1. Given that, what are the new dimensions of the output, and the new value in the bottom right corner?

Output is now 5x5, and bottom right value is $7 + 0 + 0 + 0 = 7$.

3 [15 points]. Suppose we have below 2-channel input I and filter F as follows. What is the dimensions of the output, assume padding=0 and stride=1, and what is the value in the bottom right corner of the output image?

$$I[0, :, :] = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 6 & 1 & -2 & -3 \\ -1 & 3 & 4 & 0 \end{pmatrix} \quad I[1, :, :] = \begin{pmatrix} -2 & 1 & 4 & 5 \\ 0 & 6 & 3 & 6 \\ 4 & -1 & 1 & 1 \end{pmatrix}$$

$$F[0, :, :] = \begin{pmatrix} 1 & -2 \\ 0 & 1 \end{pmatrix} \quad F[1, :, :] = \begin{pmatrix} 1 & 0 \\ 1 & -1 \end{pmatrix}$$

Output is $1 \times 2 \times 3$, and the bottom right value is $-2 + 6 + 3 + 1 - 1 = 7$.

Problem 3 [25 points] Recurrent Neural Network

Consider the following 1D RNN with no nonlinearities, a 1D hidden state, and 1D inputs u_t at each timestep. (Note: There is only a single parameter w , no bias). This RNN expresses unrolling the following recurrence relation, with hidden state h_t at unrolling step t given by:

$$h_t = w \cdot (u_t + h_{t-1})$$

The computational graph of unrolling the RNN for three timesteps is shown below, where w is the learnable weight, u_1 , u_2 , and u_3 are sequential inputs, and p , q , r , s , and t are intermediate values.

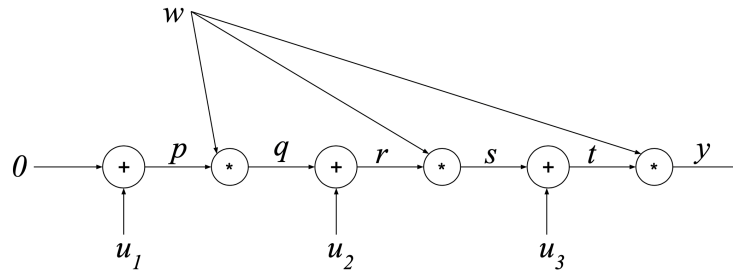


Figure 2: Computational graph in RNN for three sequential inputs

1 [8 points]. Given parameter w , and inputs u_1 , u_2 , u_3 , please write out the expression of p , q , r , s , t , and output y .

$$p = u_1$$

$$q = w \cdot p = w \cdot u_1$$

$$r = u_2 + q = u_2 + w \cdot u_1$$

$$s = w \cdot r = w \cdot u_2 + w^2 \cdot u_1$$

$$t = u_3 + s = u_3 + w \cdot u_2 + w^2 \cdot u_1$$

$$y = w \cdot t = w \cdot u_3 + w^2 \cdot u_2 + w^3 \cdot u_1$$

2 [8 points]. Why does RNN has memory and how does it achieve this function? Comparing RNN against LSTM, what is the drawback?

h_t the hidden state is updated at each time step combining the current input and the previous hidden state h_{t-1} . RNN can not handle long-term dependencies

3 [9 points]. Given the LSTM architecture as shown in Figure 3. Suppose you want the memory cell to sum its inputs over time. What values should the input gate and forget gate take?

Input = 1; forget = 1

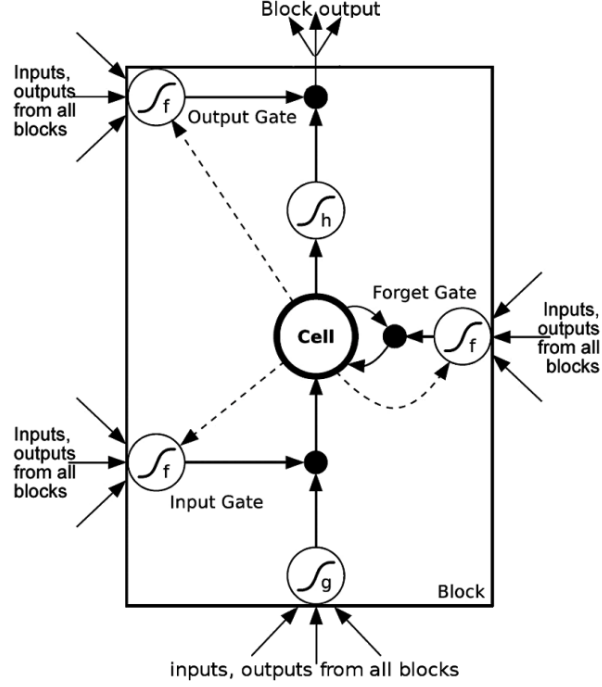


Figure 3: LSTM architecture

Problem 4 [25 points] Graph Neural Network

Consider the following undirected graph G : In this problem, we are going to work with an undirected

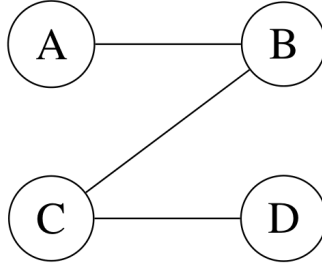


Figure 4: Graph for GNN forward pass

graph without edge weights. We will walk through a simplified forward pass of graph neural networks to help build concrete intuition for how GNNs operate. To begin with, let us assign vectors v_A, v_B, v_C, v_D to nodes A, B, C, D :

$$v_A = \begin{bmatrix} 1 \\ 2 \end{bmatrix} \quad v_B = \begin{bmatrix} -1 \\ 2 \end{bmatrix} \quad v_C = \begin{bmatrix} 2 \\ 2 \end{bmatrix} \quad v_D = \begin{bmatrix} -3 \\ 2 \end{bmatrix} \quad (10)$$

For each layer of GCN, we have the computation as

$$h_v^k = \sigma(W_k \sum_{u \in N(v) \cup v} \frac{h_u^{k-1}}{\sqrt{|N(u)||N(v)|}}), \quad (11)$$

where the activation function $\sigma(\cdot)$ is $\text{ReLU}(\cdot) = \max(0, \cdot)$. W_1 for the 1st GNN layer, and W_2 for

the 2nd GNN layer are

$$W_1 = \begin{bmatrix} 3 & 3 \\ 2 & 1 \end{bmatrix} \quad W_2 = \begin{bmatrix} 2 & 5 \\ 4 & 2 \end{bmatrix} \quad (12)$$

1 [5 points]. What is the adjacency matrix of the graph?

$$A = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

2 [8 points]. What is the first GCN layer output h_v^1 for node A, B, C, D ? (h_v^0 is the node features v).

$$h_A^1 = \sigma(W_1(\frac{v_A}{\sqrt{1 \times 1}} + \frac{v_B}{\sqrt{2 \times 1}})) = \begin{bmatrix} 11.12 \\ 4 \end{bmatrix}$$

$$h_B^1 = \sigma(W_1(\frac{v_A}{\sqrt{1 \times 2}} + \frac{v_B}{\sqrt{2 \times 2}} + \frac{v_C}{\sqrt{2 \times 2}})) = \begin{bmatrix} 13.86 \\ 5.82 \end{bmatrix}$$

$$h_C^1 = \sigma(W_1(\frac{v_B}{\sqrt{2 \times 2}} + \frac{v_C}{\sqrt{2 \times 2}} + \frac{v_D}{\sqrt{1 \times 2}})) = \begin{bmatrix} 5.37 \\ 0.17 \end{bmatrix}$$

$$h_D^1 = \sigma(W_1(\frac{v_D}{\sqrt{1 \times 1}} + \frac{v_C}{\sqrt{2 \times 1}})) = \begin{bmatrix} 5.48 \\ 0.24 \end{bmatrix}$$

3 [7 points]. What is the second GCN layer output h_v^2 for node A and B ?

$$h_A^2 = \sigma(W_2(\frac{h_A^1}{\sqrt{1 \times 1}} + \frac{h_B^1}{\sqrt{2 \times 1}})) = \begin{bmatrix} 82.45 \\ 99.94 \end{bmatrix}$$

$$h_B^2 = \sigma(W_2(\frac{h_A^1}{\sqrt{1 \times 2}} + \frac{h_B^1}{\sqrt{2 \times 2}} + \frac{h_C^1}{\sqrt{2 \times 2}})) = \begin{bmatrix} 64.11 \\ 81.59 \end{bmatrix}$$

4. [5 points] From the perspective of “message passing”, what does the k -th layer of GCN achieve in graph representation learning?

Message passing network follows an iterative scheme of updating node representations based on the aggregation from nearby nodes.